

BARONS AGCM, AB, Canada

WMO: 710070

Lat: 50.026N

Lon: 113.224W

Elev: 3173

StdP: 13.09

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-22.6	-16.0	-28.6	1.2	-22.5	-21.5	1.9	-15.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.3	87.3	60.3	83.7	59.4	80.5	58.7	65.0	78.0	63.1	76.8	61.2	75.7	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.5	88.5	71.0	58.1	81.1	68.3	55.9	75.0	65.6	31.8	77.4	30.2	76.7	28.8	76.0	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-32.4	94.2	5.8	3.8	-36.6	97.0	-39.9	99.2	-43.2	101.4	-47.4	104.2		
(5)			-32.6	69.8	5.7	2.5	-36.7	71.6	-40.1	73.0	-43.3	74.5	-47.4	76.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	19.6	18.3	29.3	39.7	50.8	57.8	63.4	62.5	54.0	41.6	29.1	17.0		
	DBStd	20.10	15.59	15.78	14.07	9.15	7.58	5.59	4.60	5.71	8.26	9.73	14.09	15.72		
	HDD50	4881	942	887	642	321	84	6	0	2	52	288	633	1024		
	HDD65	9074	1407	1307	1106	759	442	224	84	114	338	726	1078	1489		
	CDD50	1373	0	0	1	13	109	241	416	390	171	27	5	0		
	CDD65	90	0	0	0	0	2	9	35	37	7	0	0	0		
	CDH74	2737	0	0	0	18	170	274	923	980	364	8	0	0		
	CDH80	815	0	0	0	2	30	63	276	337	106	1	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	16.1	0.4	0.5	0.8	1.1	2.4	3.9	2.0	1.6	1.5	0.7	0.6	0.5		
(15) Precipitation	PrecMax	25.9	0.9	1.4	1.9	2.0	4.3	11.4	4.7	3.7	5.4	1.7	1.7	1.1		
	PrecMin	7.0	0.0	0.0	0.2	0.4	0.3	1.7	0.3	0.1	0.2	0.2	0.0	0.0		
	PrecStd	4.2	0.2	0.4	0.5	0.5	1.3	2.3	1.4	0.9	1.3	0.4	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	55.0	61.3	76.5	83.7	87.7	91.3	92.5	88.0	74.0	64.7	51.0		
		MCWB	41.8	42.8	44.9	51.5	55.2	61.9	62.6	60.9	57.4	52.0	47.5	39.5		
	2%	DB	46.8	48.3	57.1	67.9	79.0	80.7	86.6	87.6	83.1	68.2	57.0	45.6		
		MCWB	38.2	38.9	42.6	47.7	54.4	58.1	62.1	59.9	56.2	50.4	44.6	37.8		
	5%	DB	43.3	43.8	53.0	62.4	73.7	76.7	82.6	83.7	78.2	63.0	51.9	41.9		
		MCWB	36.4	35.6	40.9	44.9	52.6	57.7	61.4	59.6	55.3	47.9	42.1	35.2		
	10%	DB	40.4	39.9	48.4	57.0	68.8	73.0	79.2	79.6	72.5	58.5	47.3	38.8		
		MCWB	34.6	33.6	38.4	42.5	50.6	55.5	61.1	58.8	53.5	45.6	38.9	33.5		
	0.4%	WB	42.2	43.8	47.0	51.8	56.9	66.0	69.4	65.3	60.5	54.7	48.9	42.0		
		MCDB	51.0	53.5	58.3	73.6	76.2	78.4	80.3	80.5	78.6	70.0	62.2	49.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.8	39.3	44.0	48.5	55.2	62.1	66.3	63.4	57.9	51.5	45.8	38.3		
		MCDB	46.0	48.0	55.1	65.1	75.4	76.2	79.1	79.3	77.2	65.8	56.4	44.4		
	5%	WB	36.7	36.1	41.4	45.8	53.5	59.5	64.3	61.7	56.2	48.6	42.4	35.8		
		MCDB	42.6	42.5	51.9	60.4	71.4	72.6	76.3	78.0	74.8	61.4	51.1	41.3		
	10%	WB	34.7	33.7	38.8	43.2	51.6	57.3	62.4	60.2	54.3	46.1	39.4	33.2		
		MCDB	40.3	39.4	47.8	55.6	66.2	69.6	75.1	76.2	71.5	57.5	47.2	38.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.9	22.5	22.3	24.9	25.2	24.9	28.3	29.6	28.2	25.5	21.9	21.0		
		MCDBR	21.9	26.3	28.1	35.8	35.9	33.0	36.5	38.5	40.0	32.2	25.3	22.2		
		MCWBR	15.1	17.5	16.0	17.7	14.6	14.3	16.1	15.0	16.4	16.6	15.2	15.7		
		MCDBR	20.6	25.5	27.1	33.4	32.0	28.9	29.9	32.4	35.8	30.6	25.2	22.2		
	5% WB	MCWBR	14.6	18.0	16.0	16.8	13.5	14.0	15.3	14.3	15.2	16.3	15.5	16.3		
(40) Clear-Sky Solar Irradiance	taub		0.252	0.260	0.283	0.320	0.332	0.329	0.333	0.348	0.311	0.281	0.258	0.233		
	taud		2.442	2.482	2.480	2.387	2.404	2.460	2.462	2.401	2.481	2.534	2.497	2.450		
	Ebn at Noon		260	288	297	294	293	294	291	281	283	275	257	253		
	Edh at Noon		19	23	28	34	35	34	33	34	28	22	17	16		
(44) All-Sky Solar Radiation	RadAvg		390	705	1113	1535	1829	2022	2213	1763	1250	775	429	301		
	RadStd		31	60	88	105	143	139	112	111	109	82	36	28		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (71 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page